Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 5

Question 1

The TA would like to encourage everyone in class to prepare before coming to class. To do that, the TA wants to select students who have not prepared to answer questions in class.

The TA selects k students in each tutorial to answer the questions and wants to come up with an algorithm for selecting the students. The worst case for the TA would be the case where only the k selected students have prepared, but the other n-k students in the class did not prepare. The TA thinks that an uncooperative class can make the worst case happen every time if the algorithm for selecting the students is made known. Hence there is no choice but to hide the selection algorithm **(source code or pseudocode)** from the class. Is it true that if the TA makes the selection algorithm known, the students will be able to force the worst case to happen?

- Yes
- No

Question 1

Answer: False

If the TA uses a randomized algorithm, it does not matter if he reveals the pseudo code or source code of his algorithm to the class as long as the random numbers are secret.

For example, the TA could toss a coin in class to get the random numbers.

Question 2

The TA would like to encourage everyone in class to prepare before coming to class. To do that, the TA wants to select students who have not prepared to answer questions in class.

The TA decides to select students at random to answer one question each. Assume that there are n students and k of them have not prepared. What is the expected number of questions required for finding a student who has not prepared?

- n
- n/k
- $\frac{n}{2k}$
- k/n

Question 2

Answer: B

This is a *geometric distribution* with a probability of success of $p = k/n$. The expected value of the number of trials needed to obtain a success is $1/p = n/k$.

Question 3

The TA would like to encourage everyone in class to prepare before coming to class. To do that, the TA wants to select students who have not prepared to answer questions in class.

The TA decides to select students at random to answer questions. However, the TA is worried that not everyone in class would get to participate even though many questions will be asked throughout the semester. Assume that there are n students in the class. What is the expected number of questions that need to be asked before every student has been asked a question?

- $\Theta(\lg n)$
- $\Theta(n)$
- $\Theta(n \lg n)$
- $\Theta(n^2)$

Question 3

Answer: C

This is the coupon collector problem, covered in lecture.

Next few slides from lecture notes.

Let T_i be the number of draws used to collect the i-th coupon and T be the total number of draws.

$$T = \sum_{i=1}^{n} T_i$$

Total number of draws

$$E[T] = E\left[\sum_{i=1}^{n} T_i\right]$$

Expected value

$$= \sum_{i=1}^{n} E[T_i]$$

Linearity of expectation

T_i is the time to collect the i-th coupon after i-1 coupon has been collected.

The prob of collecting a new coupon after i-1 coupon has been collected $p_i = (n-(i-1))/n$.

T_i has a geometric distribution with expectation $1/p_i = n/(n-(i-1))$.

$$
\begin{aligned}
E[T] &= \sum_{i=1}^{n} E[T_i] \\
&= \sum_{i=1}^{n} \frac{n}{n-(i-1)} \\
&= \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{1} \\
&= n \cdot \left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n} \right) \\
&= n \cdot H_n \\
&= O(n \lg n)
\end{aligned}
$$

where H_n is the n-th harmonic number and is of size $O(\lg n)$

Question 4

Suppose we are given constants a and p where a is a positive integer and $0 < p < 1$. Assume that pivot selection can be done in constant time for quicksort such that:

Case 1: the pivot will partition the array into subarrays of size pn and $(1\text{-}p)n$ respectively.

Case 2: the pivot will partition the array into subarrays of size a and $n\text{-}a$ respectively.

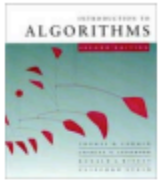
Which of the following statements are true?

- Quicksort runs in time $\Theta(n \log n)$ for both Case 1 and Case 2.
- Quicksort runs in time $\Theta(n \log n)$ for Case 1 and $\Theta(n^2)$ for Case 2.
- Quicksort runs in time $\Theta(n^2)$ for Case 1 and $\Theta(n \log n)$ for Case 2.
- Quicksort runs in time $\Theta(n^2)$ for both Case 1 and Case 2.

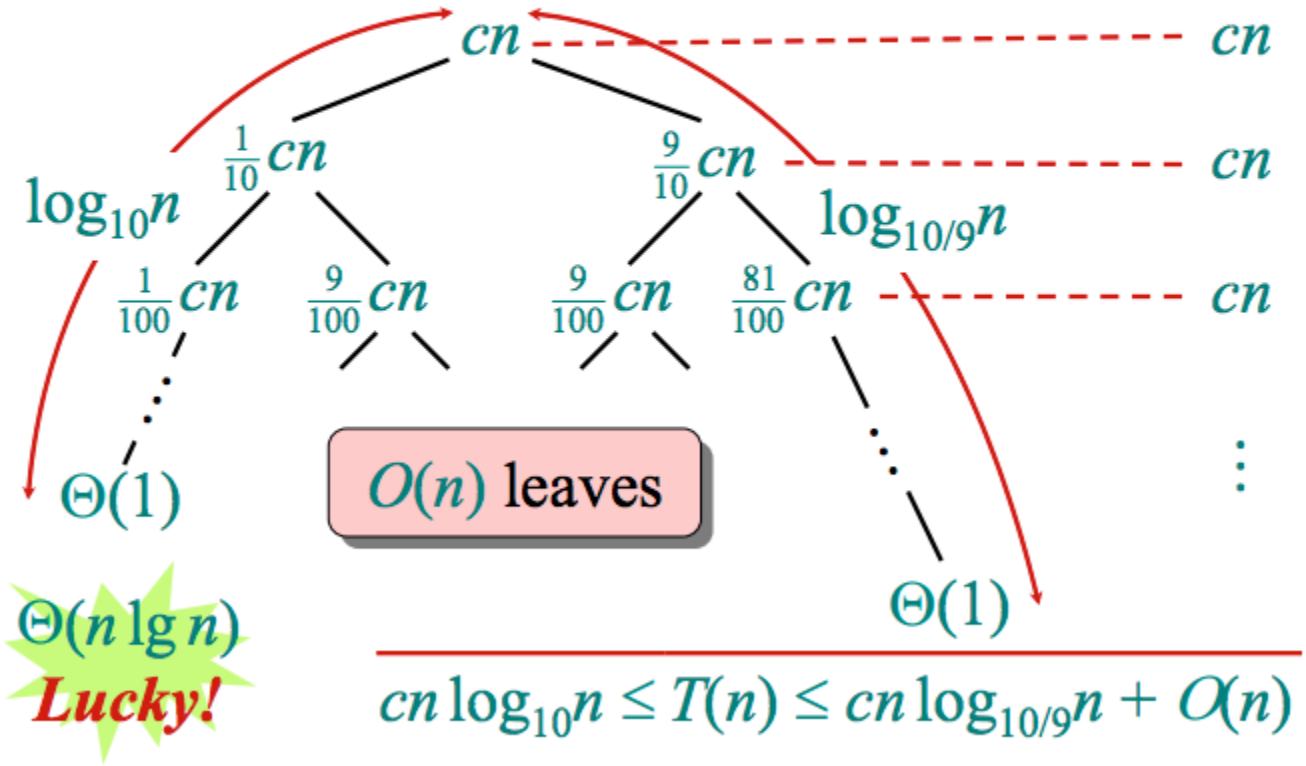
Question 4

Answer:

B.



Analysis of "almost-best" case



$$cn\log_{10}n \le T(n) \le cn\log_{10/9}n + O(n)$$

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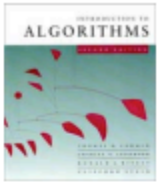
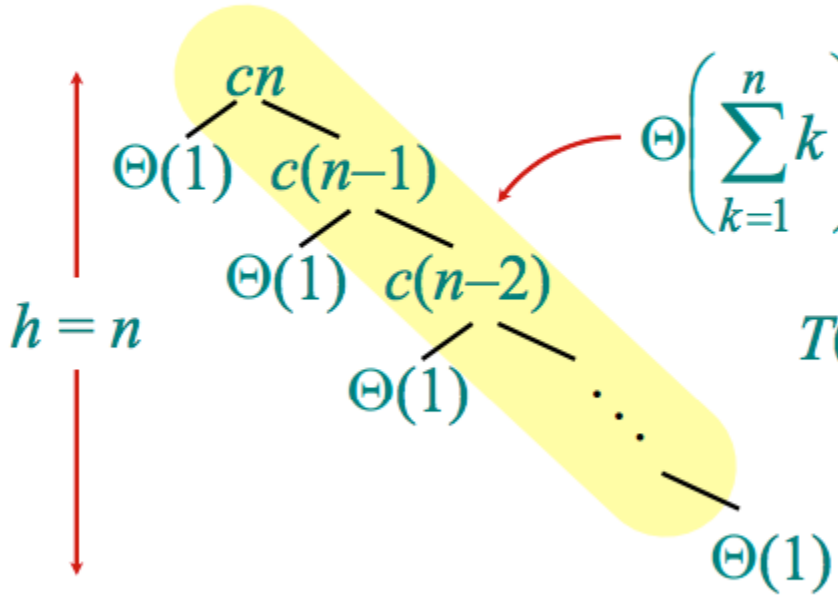
Question 4

Answer:
B.



Worst-case recursion tree

$$T(n) = T(0) + T(n-1) + cn$$



$$\Theta\left(\sum_{k=1}^{n} k\right) = \Theta(n^2)$$

$$T(n) = \Theta(n) + \Theta(n^2)$$
$$= \Theta(n^2)$$

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